Hashim Khan

Senior Mechanical Design Engineer
Hyderabad, India | +91 8840757242 | hkhan.robotics@gmail.com | linkedin.com/in/scienaut | scienaut.github.io

PROFESSIONAL SUMMARY

Senior Mechanical Design Engineer with 8+ years of experience across robotics, industrial automation, precision machine design and defence-grade engineering. Skilled in end-to-end mechanical design using Siemens NX, SolidWorks and CATIA V5, with strong application of GD&T (ASME Y14.5), tolerance stack-up analysis, engineering calculations and FEA to validate structural performance. Experienced across the full product lifecycle concept, prototyping (EVT/DVT), DFM/DFA, manufacturing (sheet metal, CNC machining, 3D printing, injection molding) and production hand-off - with solid practice in multilevel BOM creation, engineering change management and technical documentation. Comfortable integrating mechatronic components (sensors, actuators, PCBs) into electromechanical assemblies, coordinating with vendors and cross-functional teams, and mentoring junior engineers. Adaptable to new industries and standards, with a hands-on, problem-solving approach suited to fast-paced engineering environments.

CORE COMPETENCIES

Mechanical Design & CAD (Siemens NX, SolidWorks, CATIA V5) | GD&T (ASME Y14.5) & Tolerance Stack-Up | Engineering Calculations & FEA | DFM/DFA | Multilevel BOM Creation & Engineering Change Management | Sheet Metal, CNC Machining, 3D Printing & Injection Moulding | Prototyping & Validation (EVT/DVT) | Mechatronic Integration (Sensors, Actuators, PCBs) | Vendor Development & Management | Cross-Functional Collaboration | Mentoring | Technical Documentation

PROFESSIONAL EXPERIENCE

Senior Mechanical Design Engineer | Retearn Technologies Pvt. Ltd. Dec 2025 – Present
Hyderabad, India

- Designed and developed Reverse Vending Machines (RVMs) for automated recycling and material recovery, directly engineering the mechanical backbone of the Deposit Return System (DRS) infrastructure for the entire Goa state- India's first-of-its-kind deployment, built to CE/ISO and EU Machinery Directive design principles
- Engineered sheet metal enclosures and brackets with material selection, tolerance stack-up and K-factor optimization, and designed high-speed knife-edge conveyor and inline singulation systems for smooth, controlled material flow.
- Developed material segregation mechanisms and drive systems integrated with IoT sensors and AI-based computer vision for automated detection and sorting, improving sorting efficiency and throughput.
- Optimized designs to reduce part count and manufacturing cost, led RCA and assembly improvements, and managed BOM/ECN processes and vendor development to raise production from 10 to 18 units (80% increase).
- Planned and executed FAT/SAT validation to ensure performance and quality compliance, while mentoring team members and collaborating across electrical, embedded software and manufacturing teams for seamless project execution.

Senior Mechanical Engineer | OEDEC Engineering Pvt Ltd Jan 2025 – Nov 2025
Pune, India

- Developed mechanical systems for marine robotics (ROV) and Special Purpose Machines (SPMs) in Siemens NX/SolidWorks, applying tolerance stack-up and FEA (structural, thermal, vibration) to design precision transmissions and sealed enclosures for harsh operating environments.
- Selected materials, heat treatments and advanced manufacturing processes (CNC machining, welding, additive manufacturing) to achieve tight tolerances in line with ISO/CE/DIN standards.
- Applied DFM/DFA and continuous-improvement methodologies (Lean, Kaizen, 5S) while managing CAD configuration, BOMs, GD&T and tolerance-stack documentation, reducing design-to-manufacture lead time by 15%.
- Collaborated with quality assurance and suppliers on CMM, NDT and compliance inspections, supporting validation and verification documentation to European regulatory requirements.

Assistant Manager | Kalyani Group, Bharat Forge Ltd Jun 2023 – Jan 2025
Pune, India

- Led design, development, testing and validation of defence-grade barrels, breech mechanisms and precision machine tools for major artillery programmes (ATAGS 155×52, MGS 155×39, Garuda 105, SRGM 76), extensively applying GD&T, material selection and tolerance stack-up analysis.
- Designed 3D CAD models, manufacturing drawings, long machining test fixtures and inspection test rigs in Siemens NX and Solid Edge, developing MQP, ITP and tooling SOPs to ISO/DEF-STAN standards.
- Performed structural, thermo-mechanical and vibration FEA alongside ultrasonic testing, magnetic particle inspection and optical borescope visual testing to validate performance and reliability.

- Led cross-functional teams on machining operations, applying Lean, 5S, Kaizen and TPM to cut man-hours by 12% and raise production output by 40%, achieving in-house manufacturing capability through new tooling.
- Managed engineering documentation (BOMs, ECNs, inspection reports), analysed production cycle-time data for forecasting, and planned tooling CAPEX for shop-floor budgets.

Engineer R&D | Kalyani Group, Kalyani Strategic Management Services TSPL Aug 2020 – Jun 2023

Pune, India

- Led product development of vibration sensor enclosures and DAQ systems through multiple design iterations, conducting material R&D across wood, aluminium, plastic and glass to balance strength, durability and aesthetics.
- Engineered Sanjeevani, a COVID-19 SpO2-based Smart Oxygen Delivery System — designing the sensor, 3D-printed medical-grade enclosure and electro-pneumatic oxygen control unit under significant time pressure.
- Designed Special Purpose Machines (SPMs) for computer-vision-based inspection and crack detection, and developed electromechanical pick-and-place robotic cells for drive shaft and crankshaft handling.
- Sensorized legacy machines for IIoT digital connectivity, building sensor data pipelines feeding AI/ML predictive-maintenance models, and maintaining CAD data, BOMs and validation reports to CE/ISO standards.

Mechanical Designer | Beauto Systems Jan 2020 – Aug 2020

Pune, India

- Led R&D for the company's first robotics project, developing AGV and robotic arm algorithms and setting up the robotics lab.
- Designed and built robotic automation systems for food processing, applying DFM/DFA principles to ISO/CE standards and producing GD&T-based fabrication drawings in SolidWorks.

Mechanical Product Developer | Footloose Labs Apr 2019 – Aug 2019

Bengaluru, India

- Developed 5–6 DOF robotic arms with integrated drive systems for pharmacy warehouse automation, performing engineering calculations and simulations in MATLAB, Octave, SolidWorks and ANSYS.
- Coordinated vendor relationships, inventory management and prototype fabrication, running mechanical testings.

Design Engineer | Aliyance Mechatronics Aug 2017 – Mar 2019

Pune, India

- Contributed to Roboseum, Gujarat Science City's robotics museum project with the Government of Gujarat, designing and simulating robotic systems including arm manipulators, rotary platforms and XYZ gantries.
- Designed and analysed robotic parts and assemblies in SolidWorks to ISO/CE standards, producing prototypes via 3D printing, and led technical recruitment from leading engineering institutes.

KEY PROJECTS

Sanjeevani Smart Oxygen Delivery System (2021) — Engineered a safety-critical electromechanical device during the COVID-19 emergency, rapidly producing CAD, engineering calculations and enclosure design under time pressure to support a working prototype.

Artillery Barrel & Breech Mechanism Programme (2023–2025) — Owned mechanical design, FEA validation and tooling development for precision defence systems, contributing to a 40% increase in production output.

EDUCATION

Bachelor of Technology (B.Tech.), Mechanical Engineering | Dr. APJ Abdul Kalam Technical University | 2013 – 2017

CERTIFICATIONS

Certified SolidWorks Professional (CSWP) | SOLIDWORKS 2024 Advanced Part Modeling | Geometric Dimensioning & Tolerancing (GD&T) | Siemens NX & CATIA V5 (Automotive, Industrial)

TECHNICAL SKILLS

CAD/CAE: Siemens NX, SolidWorks (CSWP), Solid Edge, CATIA V5, Creo, AutoCAD, ANSYS, engineering calculations, FEA/simulation

Manufacturing: Sheet metal fabrication, CNC machining, 3D printing, injection moulding, die casting, tooling & fixtures

Engineering: GD&T (ASME Y14.5), tolerance stack-up, DFM/DFA, multilevel BOM creation, engineering change management (ECN), risk assessment

Data & Process: CAD revision control, structured document/PDMenvironments, NPI/product industrialisation, EVT/DVT validation

Standards & Quality: ASME Y14.5, ISO 9001, ISO 10218, ISO 1101, ISO 2768, BS 8888, CE, DEF-STAN

Process & Tools: SAP, Lean, 5S, Kaizen, TPM, root cause analysis, supplier development, Microsoft Office, basic Python

Languages: English (professional working proficiency)